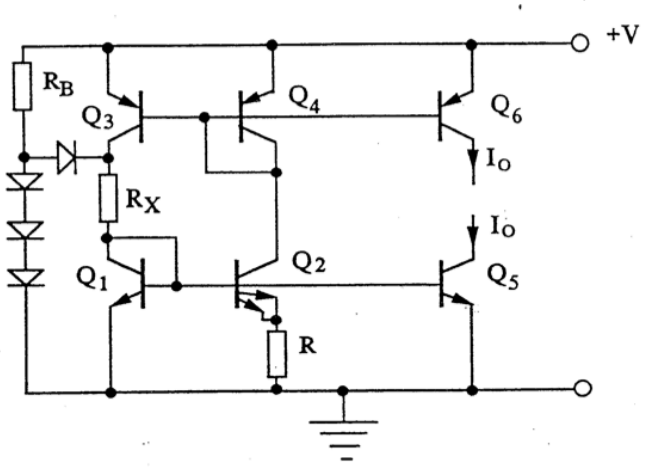
Answers to question 1:

1.a i) Self biased PTAT current source with start up circuit.



Drawing of PTAT [5]
Drawing of Startup Circuit [4]

ii) Prove that g_m is independent of temperature and therefore gain bandwidth.

$$GBW = \frac{g_{m2}}{2\pi C_c}$$

$$g_m = \frac{I_0}{V_t}$$

$$I_0 = (V_{be1} - V_{be2})/R = V_T \ln[I_{in}/I_0]/R$$

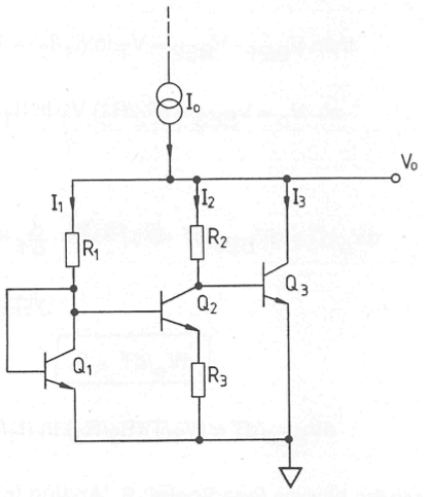
$$I_{in}/I_0 = 2$$

$$g_m = \frac{V_T \ln[2]/R}{2\pi V_t} = \frac{\ln[2]}{2\pi R}$$

$$GBW = \frac{\ln[2]}{2\pi R C_C}$$

[6]

b. Bandgap voltage reference circuit.



[5]

olutions

Answers to question 2

2.a)

$$S.R. = {I_o}/{C_C} \Rightarrow C_C = 4\,pF$$

$$GB = {g_{m2}}/{2\pi C_c} \Rightarrow g_{m2} = 7.54\times10^{-5}\ A/V$$

$$g_{m1} = g_{m2} = 2\sqrt{\beta_2 I_{D2}} \Rightarrow \beta_2 = \beta_1 = 1.42\times10^{-4} = \frac{KW}{2L} \therefore \left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_2 = 9.46 = \frac{47\mu}{5\mu}$$

$$\left(g_{o2} + g_{o4}\right) = I_{D2}\left(\lambda_N + \lambda_P\right) = 10\times10^{-6}\left[0.05\right] = 5\times10^{-7}\ \Omega^{-1}$$

$$A_1 = g_{m2}/\left(g_{o4} + g_{o2}\right) \Rightarrow A_1 = 150.8$$

$$A_2 = {10^4}/{A_1} = 66$$

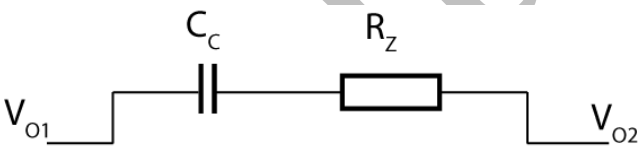
$$\left(g_{o6} + g_{o7}\right) = I_{D6}\left(\lambda_6 + \lambda_7\right) = 20\times10^{-6}\left(0.05\right) = 1\times10^{-6}\,\Omega^{-1}$$

$$A_2 = {g_{m6}}/{\left(g_{o6} + g_{o7}\right)} \Rightarrow g_{m6} = 6.63\times10^{-5}\ A/V$$

$$\Rightarrow \beta_6 = \left(\frac{g_{m6}}{2}\right)^2 \frac{1}{I_{D6}} = 5.5\times10^{-5} \Rightarrow \left(\frac{W}{L}\right)_6 = 5.5 = \frac{27\mu}{5\mu}$$

Calculation of the input pair transistor sizes [10]
Calculation of the second stage PMOS size [5]

2.b) Introducing a nulling resistor boot straps the capacitor and significantly increases the phase margin by shifting the second pole. This can also be selected such that the zero introduced cancels out the second pole.



Effectively using a resistor to try and bootstrap C_C, so that at high frequencies it breaks the feed forward path. This is achieved by making
$R_Z = 1/g_{m6}$ so that the voltage across the resistor will cancel the signal voltage on the left hand plate of the capacitor.

The location of the RHP zero however, changes to

$$Z = \frac{1}{C_C\left[\frac{1}{G_{m2}} - R_Z\right]} = \frac{1}{C_C\left[\frac{1}{g_{m6}} - R_Z\right]}$$

Or

$$f_Z = \frac{1}{2\pi C_C\left[\frac{1}{g_{m6}} - R_Z\right]}$$

Solutions

We can cancel Z and push f_Z to infinity by simply making $R_Z = \frac{1}{g_{m6}}$

[5 points]

Answers to question 3

3.a)

- An algorithmic current mode analogue-to-digital converter

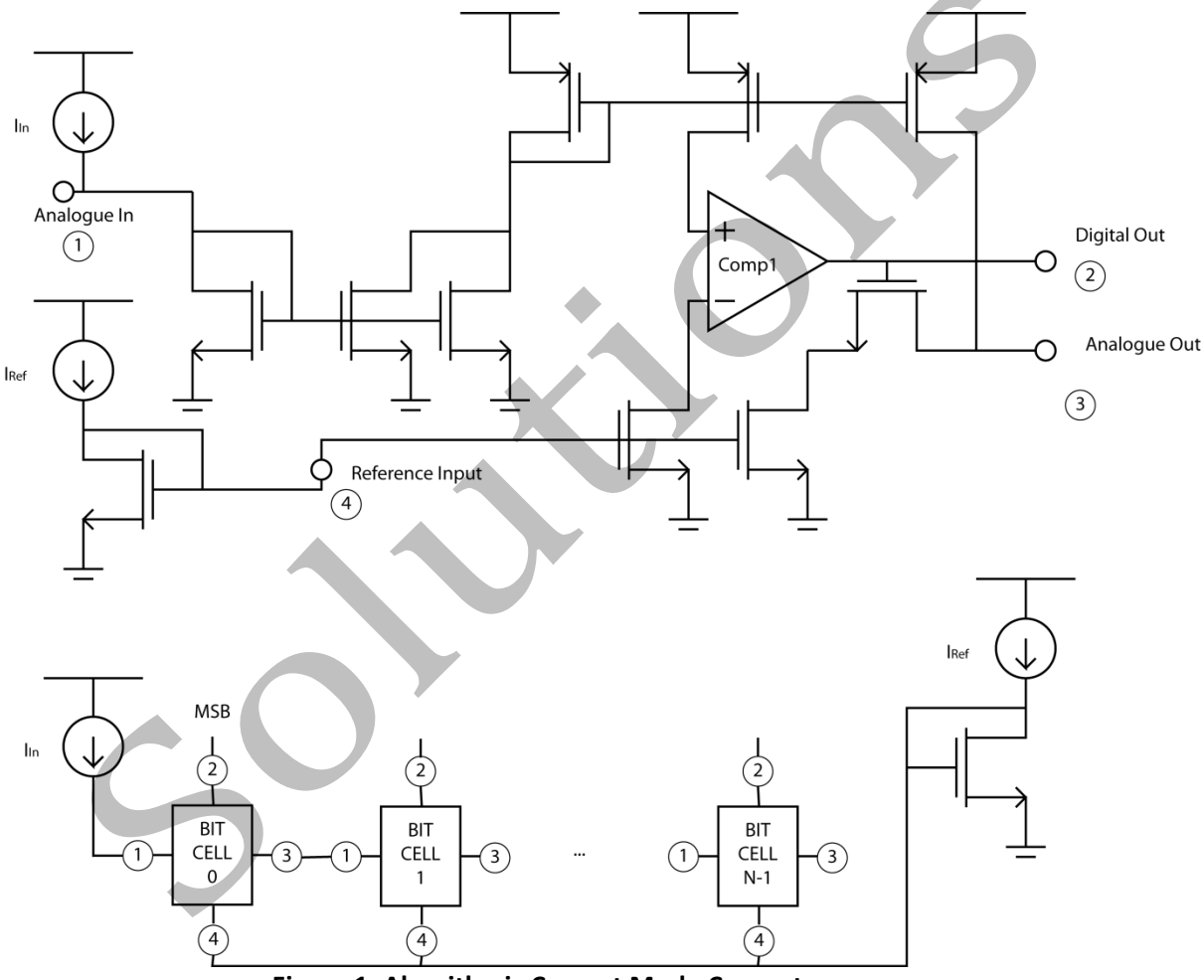


Figure 1. Algorithmic Current Mode Converter.

If 2Iin<IRef, comp goes low, digital output=0;
If 2Iin>IRef, comp output goes high, digital output=1
Analogue output ("In-IRef) feeds into following bit which performs exactly the same function. The process is repeated as many times as necessary in achieve the desired resolution.

- Sigma Delta Modulator

Solutions

In the sigma-delta modulator, coarse quantization at high sampling rate is combined with negative feedback and digital filtering to achieve increased resolution at lower sampling rates. The architecture comprises a negative feedback loop producing a coarse estimate that oscillates about the true value of input. The feedback A/D and integrator (SC) force the quantizaion error (noise) to have a high frequency spectrum. The output of the modulator is down sampled and gives a multibit digital representation oat lower sampling rate, it averages the coarse quantization leading to a much finer quantization. Since quantization noise is reduced this leads to higher resolution and S/N ratio. When N is 1 the coarse estimate and A/D converter have a 1 bit resolution and this part of the circuit is referred to the sigma-delta modulator. Basically quantisation noise is reduced because the digital filter (decimator) rejects quantisation noise above the signal. Over sampling stretches noise but keeps the signal the same. Generally quantisation noise is flat over the oversampled frequency range. Higher order sigma-delta modulators shape the quantisation noise (eg low at low frequency and high at high frequency) hence filtering will reject a higher proportion.

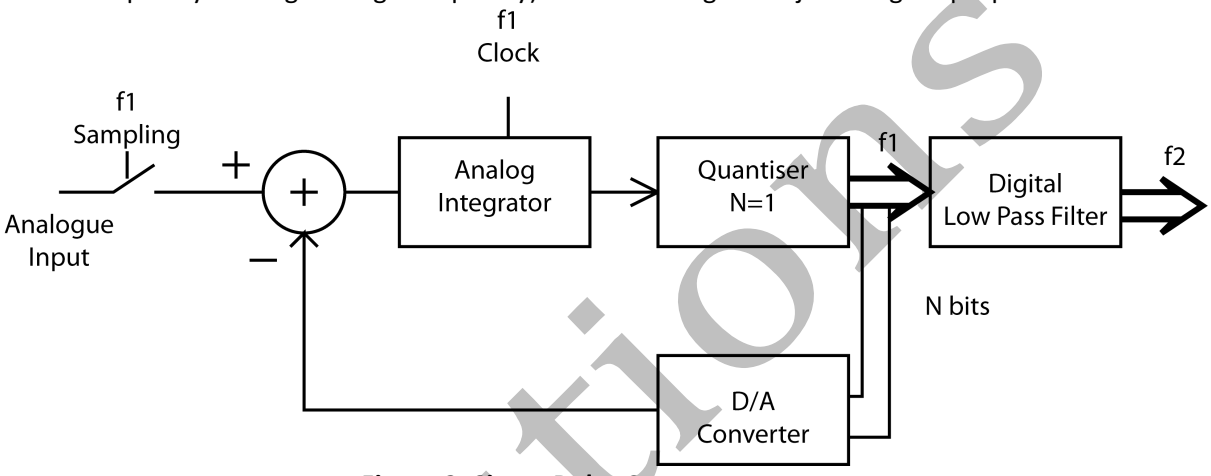


Figure 2. Sigma Delta Converter.

Describing one of the two converters and drawing it [10]

3.b)

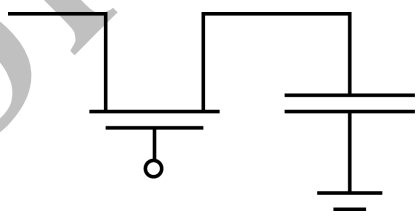


Figure 3. Sampling Switch Configuration.

$DR \therefore V_{\mathrm{Re}f} \,/\, Noise = 2^N$

RMS noise of switch capacitance $\sqrt{\frac{kT}{C}}$

Assume

$f_C = \frac{1}{10.R.C}$ Then solving for C gives

$DR = 2^N = V_{\mathrm{Re}f} \,/\, \sqrt{kT\,10Rf_C}$

$R_{On} = \frac{1}{2\beta\left(V_{gs} - V_T\right)} \approx \frac{1}{\left(2\beta \times 4\right)} = 100k\Omega \;;\; \beta = \left(\frac{kW}{2L}\right)$

Solutions

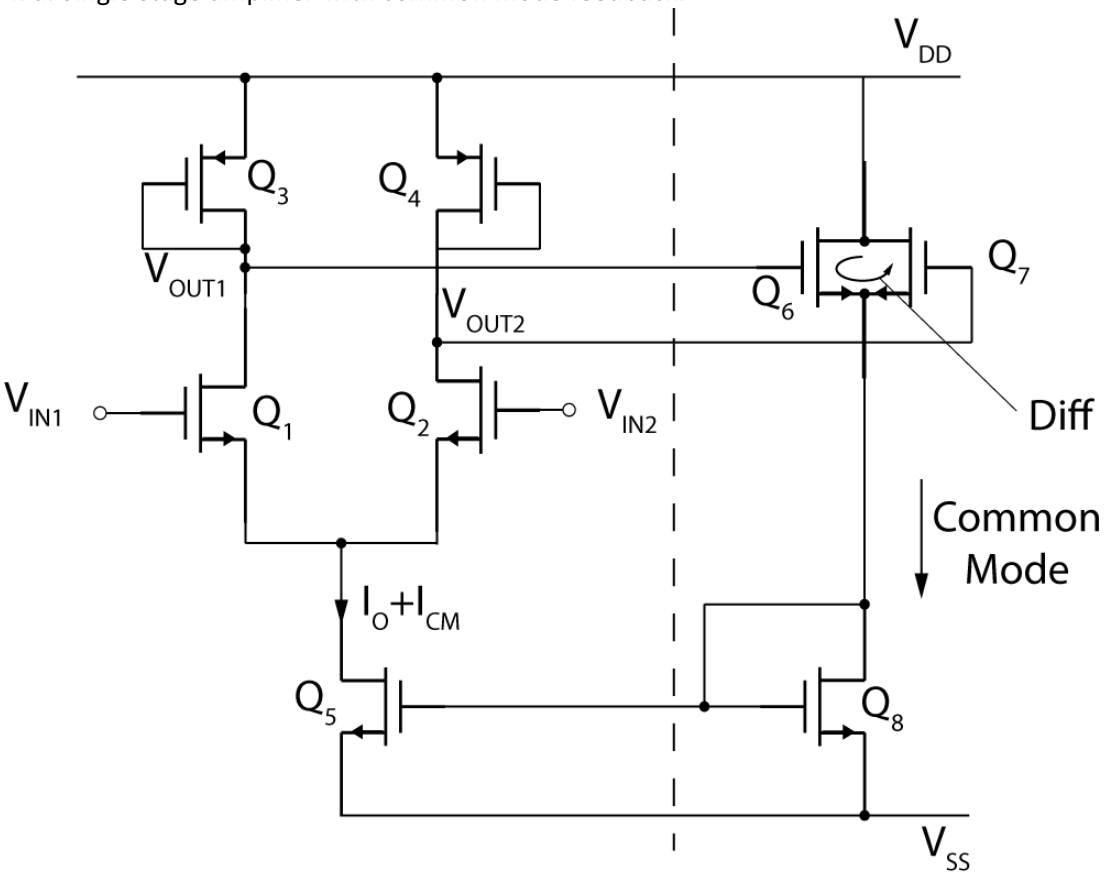
Can now find DR 40kHz,

$$DR = 3.88 \times 10^5 \Rightarrow N = 18.5 -> 18$$

Calculation of the resolution [10]

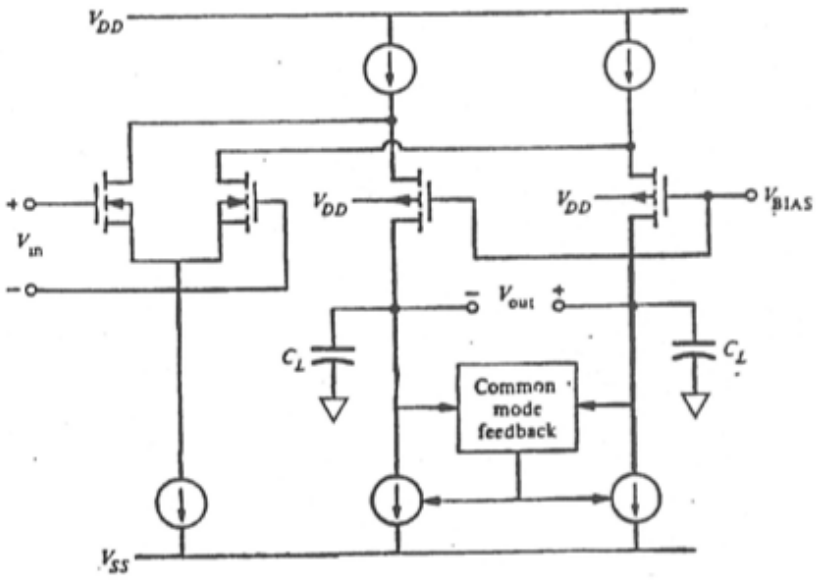
Answers to Question 4

4. a. Single stage amplifier with common mode feedback.



Folded Cascode is also an acceptable answer:

SO



.

[6]

b. Mismatches like λ (the Vds difference), threshold voltage VT and W/L may cause errors in offset and gain. A proper layout may help reduce the effect of such mismatches as well as gradients. Common centroid design splits the transistors over a common centroid or axis of symmetry. For example, the elements A and B in an array may be inter-digitated as in figure below. The input differential pair transistors in the Op-amp are the critical ones for the layout as they affect the offset and gain.

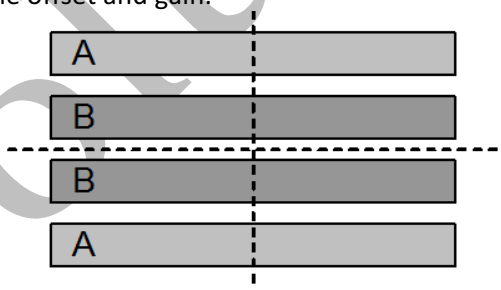


Figure 4. Question 4 part b; interdigitation over a common centroid.

Description of common centroid concept and telling the crcitical transistors of the Op-Amp for the layout [4]

c.i.) Circuit is a differential continuous time integrator with a balanced double differential linear active transresistor.

Solutions

$$I_1 = 2\beta[(V_g - V_x - V_T)(V_{in}/2 - V_x) - 1/2(V_{in}/2 - V_x)^2]$$
$$I_2 = 2\beta[(V_g - V_x - V_T)(-V_{in}/2 - V_x) - 1/2(-V_{in}/2 - V_x)^2]$$
$$R = 2V_{in}/(I_1 - I_2)$$
$$R = \frac{1}{\beta(V_g - V_T)}$$

[5]

c.ii) Time constant τ=RC=10ms
R=1/20ux5(1-0.5)=20 KΩ
C=τ/R=10m/20K=5x10^{-7} f

[5]

Answers to question 5

5.a)

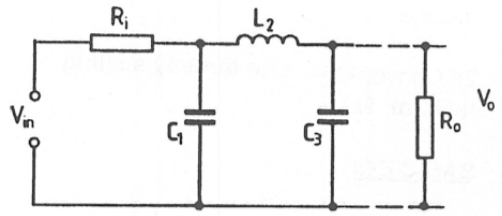


Figure 5. 3rd Order Ladder Chebyshev Filter.

[5]

5. b)

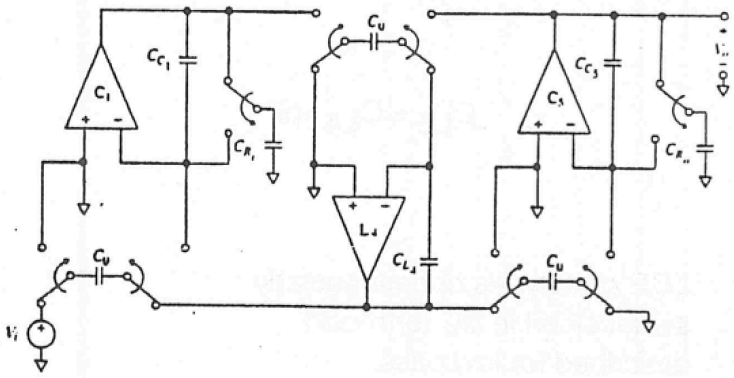


Figure 5b. 3rd-order Chebyshev Low-pass Switched-capacitor Ladder Filter.

Drawing the 3rd order Chebyshev low pass filter and its switched capacitor ladder equivalent
[5]

5. c.i

$$f_c = \frac{2\pi f_o}{L_2}\frac{C_{L2}}{C_u} = \frac{2\times 3.14\times 5\times 10^3}{1.096}\frac{3.49}{1} = 99.99kHz \approx 100kHz$$

[4]

5. cii

$$C_1 = C_3 = \frac{1.596}{2\pi f_o} = \frac{1.596}{2\times 3.14\times 5\times 10^3} = 5.08\times 10^{-5}\ F$$

$$\frac{C_{c1}}{C_u} = f_c C_1 R_s\ ; R_s = 1 \Rightarrow C_{c1} = C_{c3} = 100\times 10^3 \times 5.08\times 10^{-5}\times 10^{-12} = 5.08\,pF$$

$$L = \frac{1.096}{2\pi f_o} = 3.489\times 10^{-5}$$

$$\frac{C_{L2}}{C_u} = \frac{f_c L_2}{R_s}; R_s = 1 \Rightarrow C_{L2} = 3.489\times 10^{-5}\times 100\times 10^3 \times 10^{-12} = 3.49\,pF$$

[6]

Answers to question 6

6.a VGS multiplier:

V_{GS} Multiplier

Stolen from the V_{BE} multiplier

Circuit Analysis

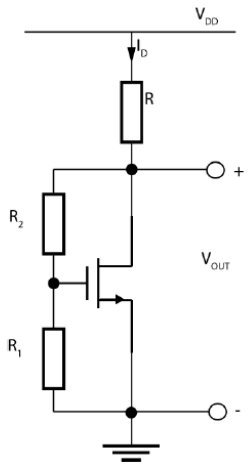


Fig. 16

$$V_{OUT} = V_{GS}\left[1+\frac{R_2}{R_1}\right]$$

OR

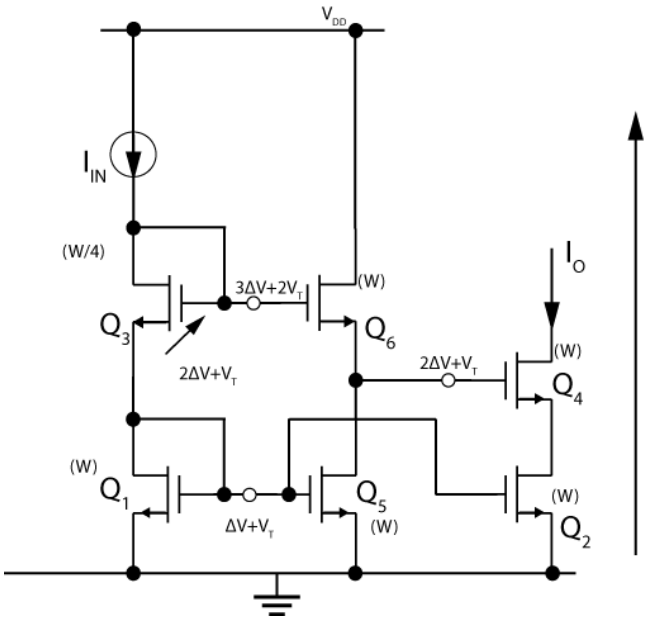
$$V_{OUT} = \left[1+\frac{R_2}{R_1}\right]\left(V_T + \sqrt{\frac{V_{DD}}{\beta R}}\right)$$

Note: All R's can be replaced with active resistors

25

[6]

b.



Assuming Equal lengths and $I_O = I_{IN}$ Remember: β = KW/2L

$$I_D = \beta\left(V_{GS} - V_T\right)^2$$

$$\beta_1 = \beta_2 = \beta_4 = \beta_5 = \beta_6 = \beta$$

$$\beta_3 = \left[\frac{\beta}{4}\right]$$

Since equal I_D, then for Q_3, $V_{DSsat} = 2(V_{GS} - V_T)$

EXAMPLE

Let $(V_{GS} - V_T) = \Delta V$ and prove to yourselves that

$$V_{OUT(\min)} = 2\Delta V$$

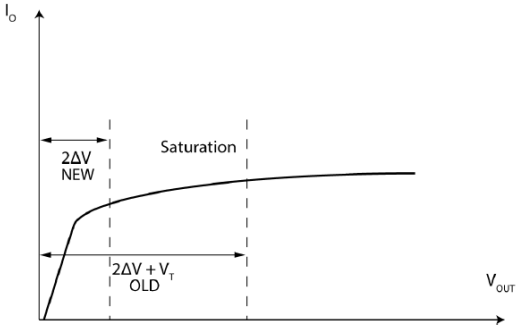


Fig. 32

HINT: *remember that $V_{DS} = V_{GS} - V_T$ or $V_D = V_G - V_T$*

52

[7]

c.b The two and four transistor potential dividers depicted in figure below.

$$K_n = 17\mu A/V^2; K_p = 8\mu A/V^2;\ V_{Tn} = V_{Tp} = 1V; V_{DD} = +5V; V_{SS} = -5V; V_{Out} = 0$$

For the two transistor potential divider

$$I_{Qp} = I_{Qn} = \tfrac{1}{2} K_n \frac{W_{Qn}}{L_{Qn}}\left(V_{out} - V_{ss} - V_{Tn}\right)^2 = \tfrac{1}{2} K_p \frac{W_{Qp}}{L_{Qp}}\left(V_{out} - V_{dd} - V_{Tp}\right)^2$$

$$\Rightarrow \frac{W_{Qn}}{L_{Qn}} = \frac{10}{136} = 0.073;\ \frac{W_{Qp}}{L_{Qp}} = \frac{10}{64} = 0.156$$

For the four transistor potential divider

$$I_{Qp1} = I_{Qn1} = I_{Qp2} = I_{Qn2} = \tfrac{1}{2} K_n \frac{W_{Qn1}}{L_{Qn1}}\left(V_{GSn1} - V_{Tn}\right)^2 = \tfrac{1}{2} K_n \frac{W_{Qn2}}{L_{Qn2}}\left(V_{GSn2} - V_{Tn}\right)^2$$

$$= \tfrac{1}{2} K_p \frac{W_{Qp1}}{L_{Qp1}}\left(V_{GSp1} - V_{Tp}\right)^2 = \tfrac{1}{2} K_p \frac{W_{Qp2}}{L_{Qp2}}\left(V_{GSp2} - V_{Tp}\right)^2$$

$$V_{GSn1} = V_{GSn2} = \frac{V_{Out} - V_{SS}}{2} = 2.5V$$

$$V_{GSp1} = V_{GSp2} = \frac{V_{Out} - V_{DD}}{2} = 2.5V$$

$$\frac{W_{n1}}{L_{n1}} = \frac{W_{n2}}{L_{n2}} = \frac{10}{19.125} = 0.523$$

$$\frac{W_{p1}}{L_{p1}} = \frac{W_{p2}}{L_{p2}} = \frac{10}{10} = 1$$

The aspect ratio of the transistors in the four-transistor potential divider is much less than the two-transistor circuit, therefore, its consuming area will be less. For example, assuming common width of 10 units, for the two transistor configuration the active area becomes 10*(64+136)=2000 units while for the four transistor it becomes 10*2*(10+19.125)=582.5 units.

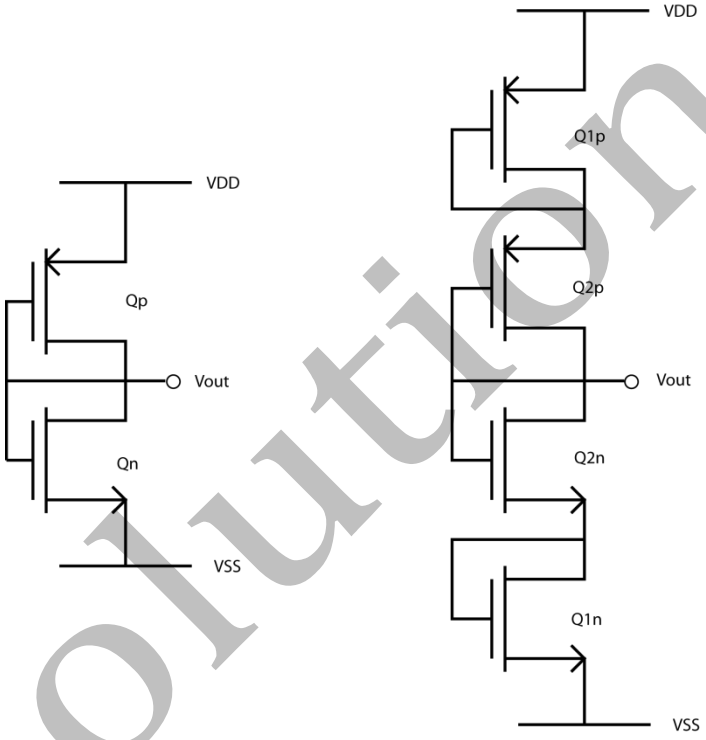


Figure 6. Answer to Question 2.b; potential Divider circuits

Drawing the potential divider with two transistor [2]
Drawing the potential divider with four transistor [2]
Calculating the required transistor sizes for each potential divider and comparing their area [3]
[Total 7 points]

Solutions